

NYCU IMF - 114 Financial Econometrics

Name: Chan Nok Hang
Student Number: 414707007

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Question 6 (abcde)

Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10		Estimation Results for Exercise 15.6			
	(1)	(2)	(3)	(4)	(5)
	OLS 1987	OLS 1988	FE	FE Robust	RE
C	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
$EXPER$	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
$EXPER^2$	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
$SOUTH$	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
$UNION$	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
N	716	716	1432	1432	1432

(standard errors in parentheses)

Figure 1: Table 15.10

- (a) The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?

Comparing OLS on 1987 and 1988

Comparison of Results:

Comparing columns (1) and (2), the OLS point estimates for the two separate years are remarkably stable. For example, the estimated return to experience ($EXPER$) is 0.1270 in 1987 and 0.1265 in 1988. The wage penalty for living in the south ($SOUTH$)

is slightly more pronounced in 1988 (-0.2384) than in 1987 (-0.2128), and the union premium (*UNION*) drops slightly from 0.1445 to 0.1102. Overall, the structural relationship appears consistent across the two cross-sections.

Heterogeneity Assumption:

When we estimate a standard cross-sectional OLS model for a single year, we are forcing any unobserved, time-invariant individual characteristics (such as innate cognitive ability, work ethic, or family background) into the general error term. The critical assumption here is that this unobserved heterogeneity is strictly uncorrelated with your explanatory variables (e.g., $Cov(x_{it}, u_i) = 0$). If individual heterogeneity is correlated with variables like union membership or experience, the OLS estimates will suffer from omitted variable bias.

- (b) The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}). \quad (XR15.6)$$

Explain any differences in assumptions between this model and the models in part (a).

Panel Data vs. Cross-Sectional Assumptions

The Decomposed Error Term:

In the panel data regression model (XR15.6), the composite error term is explicitly decomposed as $v_{it} = u_i + e_{it}$. Here, u_i represents the unobserved, time-invariant individual fixed effects, and e_{it} is the idiosyncratic error term that varies across both time and individuals.

Difference in Assumptions:

The fundamental difference lies in how u_i is treated. The OLS models in part (a) require u_i to be completely independent of the regressors to be unbiased. The panel data model—specifically when estimated using the Fixed Effects (FE) estimator—**allows for u_i to be arbitrarily correlated** with the explanatory variables (*EXPER*, *SOUTH*, *UNION*). By using a within-transformation (time-demeaning), the FE model sweeps out u_i entirely, removing the omitted variable bias caused by time-invariant unobserved heterogeneity.

- (c) Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?

Comparing Fixed Effects (FE) to OLS

When comparing the Fixed Effects (FE) estimates in column (3) to the OLS estimates

in columns (1) and (2), we can see that controlling for unobserved individual heterogeneity significantly alters the magnitude of the coefficients, and the coefficients that demonstrate the most substantial differences in economic magnitude are experience (*EXPER*) and living in the south (*SOUTH*).

1. Quantitative Evidence

The table below summarizes the percentage changes in the coefficient magnitudes when shifting from the biased OLS models to the Fixed Effects model.

Table 1: Percentage Changes in Coefficient Magnitudes (OLS vs. FE)

Variable	Change from 1987 OLS to FE	Change from 1988 OLS to FE	Direction of Effect in FE
<i>EXPER</i>	−54.72%	−54.55%	Premium drops by more than half
<i>EXPER</i> ²	+63.64%	+61.29%	Moves closer to zero (flatter curve)
<i>SOUTH</i>	−53.24%	−36.79%	Penalty becomes much more negative
<i>UNION</i>	−43.11%	−25.41%	Premium shrinks significantly

Note: For *EXPER*² and *SOUTH*, the percentages reflect the mathematical shift relative to the original coefficient. A negative percentage for *SOUTH* means the penalty deepened.

2. Economic Intuition and Theoretical Justification

These dramatic shifts occur because the cross-sectional OLS estimates suffer from severe omitted variable bias. OLS forces all unobserved, time-invariant individual characteristics (such as innate ability, intelligence, work ethic, or family background)—denoted as u_i —into the error term. The Fixed Effects model applies a within-transformation that completely sweeps out u_i , revealing the true partial effects.

The Drop in Experience (*EXPER*):

The OLS models heavily overestimate the return to experience. In reality, highly capable or motivated individuals (positive unobserved traits in u_i) are more likely to remain in the workforce longer and accumulate more experience. OLS incorrectly attributes the wage boost from this unobserved ability to the *EXPER* variable. Once FE controls for individual ability, the true return to an additional year of experience is shown to be less than half of the OLS estimate.

The Deepening of the Regional Penalty (*SOUTH*):

The OLS estimates underestimate the severity of the wage penalty for living in the South. This suggests that there are unobserved worker characteristics correlated with living in the South that actually have a positive effect on wages, masking the true negative impact of the region itself. Controlling for these fixed individual traits reveals

that the geographic penalty is substantially harsher.

The Shrinking Union Premium (*UNION*):

The union wage premium drops significantly in the FE model. This aligns with the theory of positive selection into unions: workers with higher unobserved skill levels or motivation (who would earn higher wages regardless) are often the ones who secure union jobs. FE removes this selection bias, showing that the isolated effect of unionization is much smaller than OLS implies.

In conclusion, the unobserved individual heterogeneity is highly correlated with the regressors. The Fixed Effects model eliminates this bias, causing the massive adjustments seen primarily in *EXPER* and *SOUTH*.

- (d) The F-statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F-distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.

1. The Hypotheses (from Equation 15.19)

The test evaluates whether every individual in the panel has the exact same intercept (i.e., no unobserved individual differences).

- Null Hypothesis (H_0): $\beta_{11} = \beta_{12} = \dots = \beta_{1N} = \beta_1$ (All individuals share a common intercept; pooled OLS is appropriate).
- Alternative Hypothesis (H_1): The individual intercepts (β_{1i}) are not all equal (Unobserved heterogeneity exists; Fixed Effects is required).

2. Degrees of Freedom (from Equation 15.20)

The F -statistic formula in equation (15.20) directly gives us the degrees of freedom for the numerator and the denominator.

Numerator Degrees of Freedom (df_1): This represents the number of restrictions we are placing on the model. We are restricting $N - 1$ individual intercepts to equal one common intercept.

- $df_1 = N - 1$
- $df_1 = 716 - 1 = \mathbf{715}$

Denominator Degrees of Freedom (df_2): This represents the residual degrees of freedom from the unrestricted (Fixed Effects) model. From the formula, this is $NT - N - K_S$, where K_S is the number of time-varying slope regressors (*EXPER*, $EXPER^2$, *SOUTH*, *UNION*).

- $N = 716$

- $NT = 1432$
- $K_S = 4$
- $df_2 = 1432 - 716 - 4 = \mathbf{712}$

3. The Critical Value

We are asked to find the 1% level of significance critical value for the F -distribution with 715 and 712 degrees of freedom:

$$F_{0.01}(715, 712) = 1.190496.$$

4. Conclusion

Test Statistic: The problem states the calculated F -statistic is 11.68.

Decision Rule: Reject H_0 if $F > F_{critical}$.

Econometric Conclusion: Because 11.68 is massively larger than the 1% critical value of 1.190496, we firmly reject the null hypothesis.

There is overwhelming statistical evidence that the individual intercepts (β_{1i}) are not all equal. Unobserved individual heterogeneity is present, meaning the restricted pooled OLS model is inappropriate and the Fixed Effects model should be used.

- (e) Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?

1. Difference in Assumptions

When estimating the fixed effects model without robust standard errors (Column 3), we impose the strict assumption that the idiosyncratic error term, e_{it} , is both homoskedastic (constant variance) and exhibits no serial correlation over time (i.e., $Cov(e_{i,1987}, e_{i,1988}) = 0$). Conversely, estimating the model with cluster-robust standard errors (Column 4) relaxes these constraints. By clustering the standard errors at the individual level, we allow for arbitrary heteroskedasticity and arbitrary within-cluster serial correlation. This acknowledges the econometric reality that an individual worker's unobserved wage shocks in 1987 are highly likely to be correlated with their own unobserved wage shocks in 1988.

2. Comparison of Standard Errors

While the point estimates remain mathematically identical across both models, the standard errors adjust to account for this within-cluster correlation. Comparing Column (4) to Column (3):

- The standard errors for the intercept (C), $SOUTH$, and $UNION$ increase.
- The standard error for $EXPER$ decreases very slightly (from 0.0330 to 0.0328).

- The standard error for $EXPER^2$ remains unchanged.

3. Substantial Differences and Direction of Change

The only standard error that is substantially different is the one for *SOUTH*, which nearly doubles in magnitude, increasing from 0.1258 to 0.2495.

In general, the cluster-robust standard errors in this model are larger. This aligns with standard panel data theory: positive serial correlation within individual clusters typically causes default (non-robust) standard errors to be artificially underestimated. The cluster-robust estimation corrects this downward bias, resulting in larger, more reliable standard errors.

Question 20 (abc)

This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- (a) Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?

Based on the regression output, here are the answers to the specific questions regarding student performance:

1. Do students perform better in reading when they are in small classes?

Yes. The coefficient for *small* is 5.8228 and is highly statistically significant (p-value = 4.19e-09, which is far below 0.01). This means that, holding all else constant, students in small classes score approximately 5.8228 points higher on the reading test compared to students in regular-sized classes.

2. Does a teacher's aide improve scores?

No, there is no statistical evidence for this. While the coefficient for *aide* is positive (0.8178), the p-value is 0.391. Because the p-value is much greater than standard significance levels (like 0.05 or 0.10), the result is not statistically significant. We cannot conclude that having a teacher's aide has any measurable effect on reading scores.

3. Do the students of more experienced teachers score higher on reading tests?

Yes. The coefficient for *tchexper* is 0.4925 and is highly significant (p-value = 1.61e-

12). This indicates that for every additional year of a teacher's experience, a student's reading score increases by roughly 0.49 points, holding other factors constant.

4. Does the student's sex or race make a difference?

Yes, both demographics are highly significant.

- Sex: The coefficient for boy is -6.1564 (p-value = $1.23e-14$). This means male students score approximately 6.16 points lower than female students, on average.
- Race: The coefficient for white_asian is 3.9058 (p-value = $4.26e-05$). This means that White and Asian students score about 3.91 points higher than students in the baseline race category (which likely includes Black, Hispanic, and other minority students).

5. Other Notable Findings:

The strongest predictor in the model is *freelunch*. The coefficient is -14.7713 and is overwhelmingly significant (p-value $\leq 2e-16$). Since qualifying for free lunch is a proxy for lower socioeconomic status, this shows a severe negative impact on reading scores, dropping them by nearly 15 points.

Summary

The pooled OLS model suggests that small class sizes, more experienced teachers, being female, being White/Asian, and not qualifying for free lunch all significantly improve student reading scores. Interestingly, assigning a teacher's aide to a regular classroom provides no statistically significant benefit.

- (b) Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [*Hint*: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]

Comparing OLS to School Fixed Effects

1. Direct Comparison of Estimates

First, let's put the coefficients side-by-side to see exactly what happened when we swept out the unobserved school-level heterogeneity (like school funding, principal quality, or neighborhood effects).

Table 2: Comparison of Pooled OLS and School Fixed Effects Estimates

Variable	OLS Estimate (Part a)	FE Estimate (Part b)	Change in Magnitude
<i>SMALL</i>	5.823***	6.490***	Increased
<i>AIDE</i>	0.818	0.996	Still Insignificant
<i>TCHEXPER</i>	0.492***	0.286***	Dropped significantly
<i>BOY</i>	-6.156***	-5.456***	Shrank slightly
<i>WHITE_ASIAN</i>	3.906***	8.028***	More than doubled
<i>FREELUNCH</i>	-14.771***	-14.594***	Practically unchanged

Note: *** indicates $p < 0.001$.

2. Have any of your conclusions changed?

Qualitatively, NO:

The directional conclusions and statistical significance remain exactly the same. Small classes still improve reading scores, teacher's aides still do not have a statistically significant effect, more experienced teachers still improve scores, female students score higher than males, White/Asian students score higher than the baseline, and students on free lunch score significantly lower.

Quantitatively, YES:

The magnitudes of the true effects were suffering from omitted variable bias in the pooled OLS model. Controlling for school fixed effects reveals a few major adjustments:

Teacher Experience (*tchexper*): The return to teacher experience drops from 0.492 to 0.286. This proves the OLS estimate was heavily upward-biased. Why? Better-funded or more highly-regarded schools (positive unobserved school effects) are much better at attracting and retaining highly experienced teachers. OLS incorrectly gave *tchexper* the credit for the positive impact of simply being in a "good school." Once you hold the school constant (FE), the isolated effect of a teacher's experience is much smaller.

Race (*white_asian*): The coefficient jumps dramatically from 3.906 to 8.028. This suggests that White and Asian students might be disproportionately clustered in schools with negative unobserved effects (or conversely, the baseline race category is clustered in schools with positive unobservables in this specific dataset). Holding the school environment constant reveals the demographic gap is actually much wider than OLS

suggested.

- (c) Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?

1. Testing the Significance of School Fixed Effects

The F -test evaluates the null hypothesis (H_0) that all school-specific intercepts are equal to zero (i.e., there are no unobserved school effects, and the pooled OLS model is adequate).

- Test Statistic: The calculated F -statistic is 16.698 with 78 and 5681 degrees of freedom.
- P-value: $< 2.2 \times 10^{-16}$.
- Conclusion: Because the p-value is practically zero (vastly smaller than 0.01 or 0.05), we firmly reject the null hypothesis. There is overwhelming statistical evidence of significant unobserved school-level heterogeneity. Therefore, the Fixed Effects model is strongly preferred over the standard pooled OLS model.

2. Theoretical Condition for Unchanged Coefficients

For the slope coefficients to remain unchanged between OLS and FE, the unobserved school-level heterogeneity (u_i) must be strictly uncorrelated with the observed explanatory variables (X_{it}). Mathematically, this condition is:

$$Cov(X_{it}, u_i) = 0.$$

Economic Intuition in the STAR Context:

Even if schools matter significantly for reading scores (meaning the fixed effects are large and significant), OLS will only be biased if those school effects are correlated with the regressors. If the STAR experiment successfully executed perfect random assignment—meaning school quality or funding (u_i) had absolutely no correlation with class sizes (*SMALL*), teacher assignments (*TCHEXPER*), or student demographics—then OLS would actually remain unbiased. In that perfectly randomized scenario, including the school fixed effects would simply absorb residual variance and improve standard errors, but it would not significantly shift the coefficient estimates themselves.

Since the coefficients for variables like *TCHEXPER* and *WHITE_ASIAN* did shift significantly in part b, it tells us this **strict zero-correlation condition did not hold** perfectly in the data.